

Elbow placement for a redundant teleoperated arm: a nullspace posture regulator

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Abstract

A 7-DOF arm under Cartesian impedance control leaves one degree of freedom unspecified: the elbow’s position on its orbit around the shoulder–wrist axis. In a VR teleoperation system that tracks only the operator’s hand, nothing in the command stream says where that elbow should be, and a fixed joint-space pull toward the home configuration lets it drift into joint limits as soon as the wrist leaves the home region. This note reviews how the elbow is resolved in the literature—measured with an upper-arm sensor, computed analytically for spherical–revolute–spherical arms, predicted by a heuristic from the hand pose, or learned from demonstrations—and describes the solution now running in our simulator: a one-dimensional search along the self-motion orbit that treats joint-limit margin as a hard bound and a human-like swivel angle as a preference, feeding the existing nullspace spring. It needs no additional sensor, no change to the wire format, and costs about 70 μ s per tick. We give the geometry and the algorithm, report the offline validation on the FR3 kinematic chain, and list what it does not do.

1 Problem

The robot side of the system tracks the operator’s wrist with a task-space impedance law at 1 kHz,

$$\boldsymbol{\tau} = J^T(K_p \mathbf{e} - K_d J\dot{\mathbf{q}} + \mathbf{F}_{ff}) + N\boldsymbol{\tau}_{\text{null}} + \boldsymbol{\tau}_{\text{cor}} + \boldsymbol{\tau}_{\text{lim}}, \quad N = I - J^\# J, \quad (1)$$

where $J \in \mathbb{R}^{6 \times 7}$ is the end-effector Jacobian, $J^\#$ the dynamically consistent pseudo-inverse and N the nullspace projector [1]. The wrist pose fixes six of the seven joints; the seventh moves the elbow on a closed curve around the shoulder–wrist axis (Fig. 2) without moving the wrist. Whatever we put into $\boldsymbol{\tau}_{\text{null}}$ decides that curve parameter.

The previous choice was the classical gradient-projection term [2], $\boldsymbol{\tau}_{\text{null}} = K_n(\mathbf{q}_0 - \mathbf{q}) - D_n\dot{\mathbf{q}}$ with $\mathbf{q}_0$ the home configuration. This works while the wrist is near home and fails otherwise: the pull toward $\mathbf{q}_0$ is meaningless 30 cm away from it, so the elbow wanders, joint limits are hit, and the joint-limit torque $\boldsymbol{\tau}_{\text{lim}}$ has to fight the posture term. Two things are wanted instead. First, the arm should stay inside its joint-limit margins on its own. Second, among the configurations that do, the elbow should be where the operator expects it—roughly where their own elbow is—so that self-collision avoidance and camera occlusion become intuitive rather than surprising.

2 How others resolve the elbow

The literature on human-like redundancy resolution is large (see [8] for a review); for our purpose it sorts by what input the method needs (Fig. 1).

Measured elbow. The most direct route is to measure the operator’s elbow and reproduce it. OBRAM [9] captures shoulder, elbow and wrist with a Vicon system and solves one bounded nonlinear least-squares per frame in which the wrist pose is the functional term and the normalised arm-plane normal $\mathbf{n} = \mathbf{v}_{se} \times \mathbf{v}_{ew} / \|\cdot\|$ is the anthropomorphic term; it reaches sub-degree swivel error at about 38 Hz and explicitly avoids nullspace projection so that a single joint solution can feed a joint impedance controller. Wearable variants replace the cameras with magneto-inertial units on the upper arm and forearm [10]. Whole-body humanoid teleoperation systems such as OmniH2O [12] retarget full mocap keypoints, elbow included, and let the tracking policy sort out the rest. These methods are exact by construction, but each requires a sensor we do not have: our operator carries a headset and two wands, tracked by one Lighthouse, and a third tracker on the upper arm would be occluded whenever the arm crosses the body.

Analytical. For spherical–revolute–spherical (SRS) arms the redundancy is a single scalar, the swivel or “arm” angle, and inverse kinematics can be closed-form in it [3, 4, 5]. Shimizu et al. additionally map joint

How the redundant elbow DOF is resolved

grouped by the input each method needs

Measured elbow	Analytical	Heuristic / regression	Learned
upper-arm tracker, IMU or mocap marker OBRAM [9], M-IMU [10], whole-body mocap [12] exact; needs hardware	swivel / arm-angle IK for SRS arms [3], [4], [5] closed form; still needs the angle from somewhere	elbow from hand pose [6], [7], VR avatar IK no sensor; approximate; operator-independent	network predicts elbow or SEW angle from EE pose [13], [14] needs demonstrations; operator-specific
This work: nullspace posture regulator heuristic swivel target (from the home pose) as preference, joint-limit margin as hard bound, 1-D search on the orbit; no sensor, no wire-format change, $\sim 70 \mu\text{s}$ per tick, sits under the existing Cartesian impedance controller			

Figure 1: Families of elbow resolution, grouped by the information they consume. Our method sits in the heuristic family but keeps feasibility as a hard constraint and needs only the joint state the controller already has.

limits onto feasible intervals of the arm angle, which is the analytical ancestor of the interval logic we use below. The FR3 is not SRS (the 88 mm and 82.5 mm offsets break the spherical wrist and elbow), so these closed forms do not apply directly; REMTC [11] handles that by solving an SRS approximation first and refining numerically, which OBRAM reports as a source of jitter. Analytical IK also only moves the problem: it still needs the angle from somewhere.

Heuristic and regression. A second line predicts the elbow from the hand pose alone, exploiting the regularities of human arm posture. Kim, Li, Milutinović and Rosen [6] model the swivel angle as the one that would let the palm face the head—a “virtual target” criterion—and validate it against exoskeleton data; Zanchettin, Bascetta and Rocco [7] fit a linear regression of the swivel angle on hand position and show that robots resolved this way are perceived as more acceptable by nearby humans. VR avatar systems use similar closed-form heuristics to place an elbow from hand and head poses at frame rate. These are sensor-free and cheap; their weakness is that they describe a typical arm, not this operator, and cannot follow a deliberate elbow lift.

Learned. Recent work replaces the hand-crafted regression with a network. HL-IK [13] predicts the elbow pose from the target end-effector pose and a short history with a FiLM-modulated spatio-temporal attention network and adds it as a weighted residual in a Levenberg–Marquardt IK, cutting arm-similarity error by 30–47% at 2 ms overhead; Shi et al. [14] learn operator-specific shoulder–elbow–wrist angles from exoskeleton demonstrations and feed them to a differential IK, reporting sub-millisecond inference. These improve over heuristics exactly where the heuristic is wrong—the individual operator—but need demonstrations, ideally with an elbow sensor, which is the hardware we are trying to avoid. They are a natural upgrade path once a tracker is borrowed for a few sessions.

Where the control law lives. Orthogonal to the choice of target is whether the posture is imposed as a full IK solution (OBRAM, HL-IK, REMTC), which then requires a joint-space tracking controller, or as a secondary task in the nullspace of a task-space controller [2, 1]. Our arm control is Cartesian impedance for good reasons (compliance in contact, no IK discontinuities between the operator’s hand and the robot), so the posture has to enter through $N\tau_{\text{null}}$, and the question reduces to *which joint reference* $\mathbf{q}_{\text{ref}}$ that spring should pull toward.

Elbow orbit and swivel angle

wrist held by the task-space impedance; $\mathbf{p}_e$ is free on the dashed orbit; $\varphi = \text{atan2}(\mathbf{e} \cdot \mathbf{t}, \mathbf{e} \cdot \mathbf{r})$

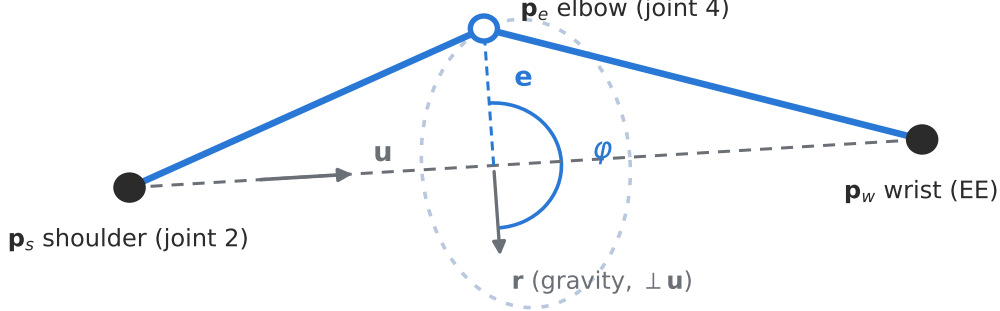


Figure 2: Shoulder $\mathbf{p}_s$ (joint-2 origin), elbow $\mathbf{p}_e$ (joint-4 origin) and wrist $\mathbf{p}_w$ (end effector). With the wrist held, the elbow moves on the dashed orbit; its position is the swivel angle φ , measured in the plane perpendicular to the shoulder–wrist axis $\mathbf{u}$ from the projected gravity direction $\mathbf{r}$.

3 Our method

3.1 Geometry

Let $\mathbf{p}_s, \mathbf{p}_e, \mathbf{p}_w$ be the shoulder, elbow and wrist positions in the base frame, obtained from forward kinematics of the joint-2, joint-4 and end-effector frames (libfranka’s `Model::pose` on hardware, Pinocchio in simulation, behind one functor). Define

$$\mathbf{u} = \frac{\mathbf{p}_w - \mathbf{p}_s}{\|\mathbf{p}_w - \mathbf{p}_s\|}, \quad \mathbf{e} = \frac{(I - \mathbf{u}\mathbf{u}^\top)(\mathbf{p}_e - \mathbf{p}_s)}{\|\cdot\|}, \quad \mathbf{r} = \frac{(I - \mathbf{u}\mathbf{u}^\top)\mathbf{g}}{\|\cdot\|}, \quad \mathbf{t} = \mathbf{u} \times \mathbf{r}, \quad (2)$$

with $\mathbf{g}$ the world “down” direction expressed in the base frame (the FR3 bases are wall-mounted, so this is not the base $-z$). The swivel angle is

$$\varphi(\mathbf{q}) = \text{atan2}(\mathbf{e} \cdot \mathbf{t}, \mathbf{e} \cdot \mathbf{r}), \quad (3)$$

and a target angle φ^* defines a target elbow direction $\mathbf{e}^* = \cos \varphi^* \mathbf{r} + \sin \varphi^* \mathbf{t}$. The preference cost is the direction-cosine mismatch

$$c(\mathbf{q}) = 1 - \mathbf{e}(\mathbf{q}) \cdot \mathbf{e}^*, \quad (4)$$

which is smooth and free of angle wrapping. The same quantity underlies OBRAM’s arm-plane normal (the normal is $\mathbf{u} \times \mathbf{e}$ up to sign), so the two criteria are equivalent for a fixed wrist.

The target. Without an elbow sensor the target is a heuristic in the spirit of [6, 7]:

$$\varphi^* = \varphi(\mathbf{q}_0) + \Delta\varphi_0 + k_h(z_w - z_{w,0}) + k_l(y_w - y_{w,0}), \quad (5)$$

where $\varphi(\mathbf{q}_0)$ is measured from the home configuration at start-up and the height/lateral modulation gains are currently zero. Two consequences are deliberate. The home posture is the neutral elbow by construction, so nothing moves after homing, and the two arms with mirrored home poses get mirrored targets without any hand-written sign. And because $\mathbf{e}$ is a direction, the target is invariant to where the wrist is and to the hand–gripper orientation offset that accumulates over clutches; the elbow reference does not need re-anchoring when the operator clutches.

3.2 Algorithm

Each state-thread tick (200 Hz) the regulator receives the measured $\mathbf{q}$ and runs the steps of Fig. 3.

Tangent. The self-motion direction is the null vector of $J(\mathbf{q})$, taken as the last right-singular vector $\mathbf{v}$ of its SVD. Its sign is arbitrary between ticks and is flipped to satisfy $\mathbf{v} \cdot \mathbf{v}_{\text{prev}} > 0$.

Per state tick (200 Hz), $\sim 70 \mu\text{s}$

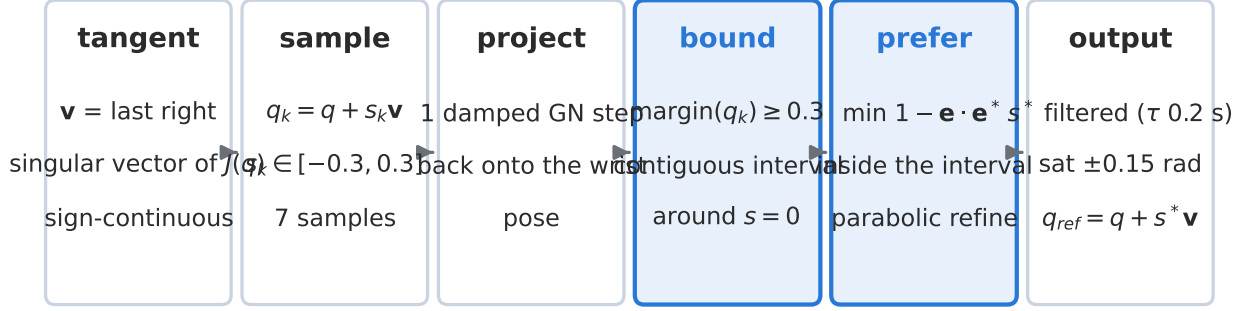


Figure 3: One update. Highlighted steps are the two decisions; the rest is bookkeeping.

Samples on the manifold. Seven candidates $\mathbf{q}_k = \mathbf{q} + s_k \mathbf{v}$ with s_k uniform in $[-0.3, 0.3]$ rad. A straight step along $\mathbf{v}$ leaves the self-motion manifold at second order—up to a centimetre of wrist displacement at the ends of the window—so each sample receives one damped Gauss–Newton correction back to the current wrist pose,

$$\mathbf{q}_k \leftarrow \mathbf{q}_k - \mathbf{J}^\top (\mathbf{J} \mathbf{J}^\top + \lambda^2 \mathbf{I})^{-1} (\mathbf{x}(\mathbf{q}_k) - \mathbf{x}(\mathbf{q})), \quad (6)$$

after which the candidates are “same wrist, different elbow” to first order and the cost is evaluated where the impedance will actually hold the arm.

Feasibility as a bound. A sample is feasible if $\min_i \min(q_{k,i} - q_{\min,i}, q_{\max,i} - q_{k,i}) \geq m$ with $m = 0.3$ rad, slightly larger than the 0.25 rad at which the joint-limit torque starts, so the posture acts before the torque has to. Optionally a manipulability floor $\sqrt{\det \mathbf{J} \mathbf{J}^\top} \geq w_{\min}$ can be added; on the FR3 the elbow-straight and wrist singularities coincide with joint limits, so it is off. The feasible set used is the *contiguous* run of feasible samples containing $s = 0$. This is the basin hysteresis: when the cost is bimodal (elbow up versus elbow down are both feasible) the regulator stays in the basin it is in.

Preference inside the bound. Within the interval, the sample with the smallest $c(\mathbf{q}_k)$ is chosen and refined by a parabola through its neighbours, clamped to the interval. If the centre sample is itself infeasible—the arm is already inside the margin—the cost is ignored and the step goes toward the sample with the largest margin. Feasibility and preference are therefore never traded against each other, which is the difference from a weighted objective such as OBRAM’s $w_f O_f + w_a O_a$.

Output. The chosen displacement is low-pass filtered ($\tau = 0.2 \text{ s}$) and saturated at ± 0.15 rad,

$$\mathbf{q}_{\text{ref}} = \mathbf{q} + \text{sat}_{\pm 0.15}(\bar{s}^*) \mathbf{v}, \quad \boldsymbol{\tau}_{\text{null}} = \mathbf{K}_n (\mathbf{q}_{\text{ref}} - \mathbf{q}) - \mathbf{D}_n \dot{\mathbf{q}}. \quad (7)$$

The saturation is the speed limit: the nullspace torque can never exceed $0.15 K_n$ (3 Nm per joint at the configured gains), so the elbow drifts toward the optimum at a rate set by the existing K_n/D_n dynamics and a wrong target can never yank the arm. The reference is a pure function of $(\mathbf{q}, \text{basin})$ —no task phase, no clock—so it runs identically in the AWAITING and ENGAGED states and is reproducible offline from a joint log. Every entry into a Cartesian-control state, including after fault recovery, re-seeds $\mathbf{q}_{\text{ref}} = \mathbf{q}$.

3.3 Why this and not the alternatives

The choice follows from three constraints of the system rather than from novelty. The controller is Cartesian impedance and must stay so, which rules out full-ik schemes and puts the posture in the nullspace. The redundancy is one-dimensional, which makes a direct search along the orbit cheaper and more transparent than a QP: hard constraints become an interval, the preference is a scalar function on it, and the whole thing costs seven forward-kinematics evaluations. And there is no elbow sensor, which rules out the measured-elbow family and, for now, the learned family; the heuristic target is the honest option, and the code is written so that a regression or a network replacing φ^* is a one-function change.

4 Validation

The regulator was compiled against Pinocchio with the FR3 joint placements dumped from the project URDF and driven in a kinematic loop (Fig. 4). At the home pose the reference stays put ($\|\mathbf{q}_{\text{ref}} - \mathbf{q}_0\| = 10^{-4}$). From

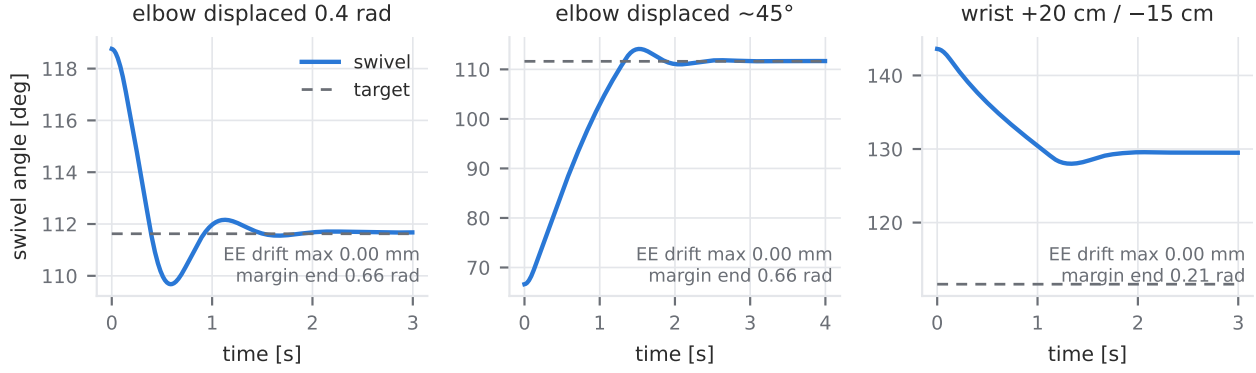


Figure 4: Offline closed loop on the FR3 chain (arm_left home pose and base orientation; kinematic proxy for the nullspace dynamics, $k = 8 \text{ s}^{-1}$, wrist held exactly). Left: elbow displaced 0.4 rad along the orbit returns to the home swivel. Middle: a 45° displacement. Right: wrist moved 20 cm forward and 15 cm down, where the whole reachable orbit has joint-6 margin below 0.21 rad; the regulator parks at maximum margin and the swivel target (dashed) is correctly ignored.

a 0.4 rad and from a 45° elbow displacement it converges back to the home swivel angle, 111.6° , with a maximum end-effector drift of 0.000 mm because the wrist is held exactly by the proxy. In the joint-limited pose it does what the design says: feasibility first. One update costs $70 \mu\text{s}$. The mild overshoot in the traces belongs to the undamped kinematic proxy, not to the regulator; on the robot D_n damps it. The same code has since run on the desk simulator in teleoperation, where the elbow behaved as intended and no posture-related faults occurred (the one fault of that session was a contact impulse unrelated to posture).

5 Limitations and next steps

The target is a typical arm, not the operator's. It reproduces a plausible elbow-down posture and keeps it consistent, but it cannot follow a deliberate elbow lift and it does not adapt to individuals; the modulation gains k_h, k_l are placeholders for a regression that needs a few sessions with a borrowed upper-arm tracker, after which a learned target in the style of [14] becomes possible without changing the controller. Feasibility wins absolutely, so in joint-limited wrist poses the elbow is not human-like at all; the margin m trades posture quality against how early the regulator acts. The search is local: seven samples over ± 0.3 rad with basin hysteresis find the nearest good elbow, never the globally best, and an arm that starts in the wrong basin stays there until the basins merge. Nullspace projection is exact only for the model, so on hardware a few millimetres of self-motion leak into the wrist; the saturation bounds the rate of that leakage but a phase-gated freeze during insertion remains an option. Near a wrist singularity the null vector is ill-defined and the posture is not meaningful there, although the filter and saturation keep the effect to a slow bounded drift. Finally, the criterion is purely kinematic: it knows nothing about the other arm, the table or the camera, which remain the job of the self-collision module and the workspace clamp.

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